

User Experience & Activation Strategy

Who uses the park, when, and why the programme targets late afternoon in spring and autumn.

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VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/
Produced by [src/models.py](#) · [src/climate.py](#) · [data/processed/hourly_climate_comfort_8760.csv](#)
Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

THE CODE AND DATA BEHIND THIS DOCUMENT

[src/models.py](#)

[src/climate.py](#)

[data/processed/hourly_climate_comfort_8760.csv](#)

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings          # section, elevation, planting
python -m src.boards            # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 38 checks
```

User Experience & Activation Strategy

Who uses this park, when, and what the climate permits

Activation strategies usually assert a programme. This one derives it. The comfort model says exactly when the park is pleasant, and the programme is built to match those windows rather than to fill a calendar — which means committing, in writing, to not programming summer midday outdoors.

Upload slot 07 — User Experience & Activation Strategy

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai

Mohamed Wasim · Individual Applicant

Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The catchment

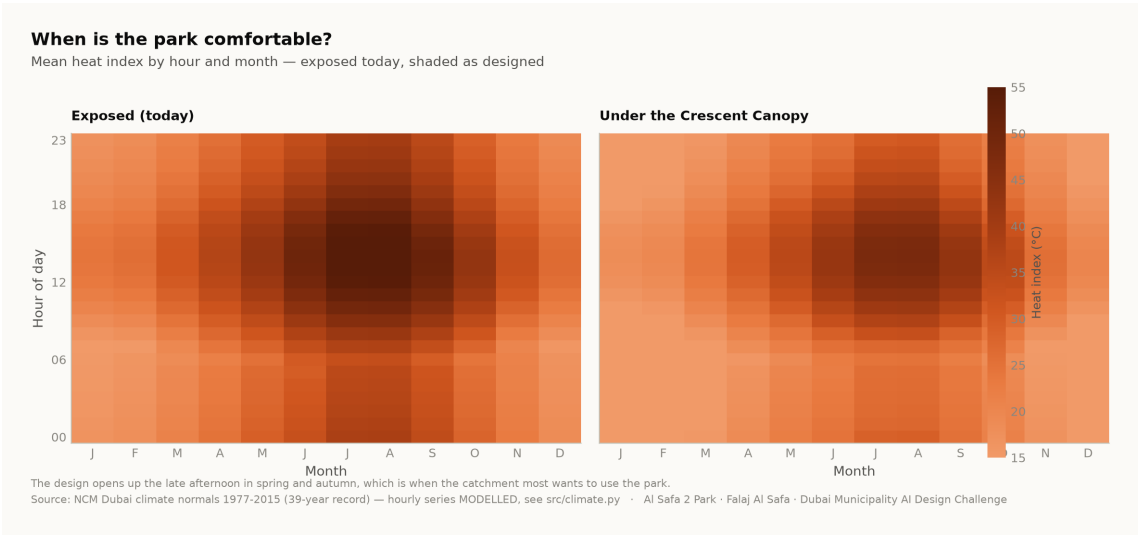
Approximately **7,640 residents** live within a ten-minute walk of the site (Dubai Statistics Center, 2023). Al Safa 2 is an established low-rise residential neighbourhood with a mixed demographic — families with young children, working adults, domestic staff, and a significant population of older residents.

2. Five user groups, and what each one needs

User group	When they come	What the design owes them
Families with young children	Late afternoon, weekends	Play in shade, seating with sightlines to it, a drinking fountain and a restroom within sight of the play area
Older residents	Early morning, evening	Step-free routes, shaded rest never far from the walk, seating at close intervals, no level change without a ramp
Runners and walkers	Dawn and after dark	A continuous loop with a known distance, lit at ankle height, separated from play and picnic
Workers on a break	Midday, year round	The one part of the site that is genuinely cool at noon — the crescent walk and the basin
Community and events	Evenings, cooler months	A flexible plaza and event lawn on the convex face, with the souk adjacent so trade and programme reinforce each other

3. When the park is actually comfortable

The comfort model resolves every hour of the year. Today 44.5% of daylight hours are comfortable; as designed, 64.6% are. The gain is not spread evenly, and that is the useful part: it concentrates in **late afternoon, in spring and autumn**.



Comfort by hour and by month. The design opens up the late afternoon in the shoulder seasons. Analysis output — computed from project data.

4. The activation calendar, derived from that

Season	Window	Programme
Oct – Apr (shoulder and winter)	16:00 – 21:00, and weekend mornings	Community events on the plaza and event lawn, evening souk trading, outdoor fitness classes, weekend family programming

Season	Window	Programme
May – Sep (summer)	Before 09:00 and after 18:00 only	Early running on Al Madar, evening use of the basin and the crescent walk, night programming on the plaza. Nothing is scheduled outdoors at midday.

Committing to a quiet summer midday is a design position, not a gap. Programming an outdoor event into a 56 °C heat index would be a scheduling decision that the analysis in this submission directly contradicts. The park is designed so that the hours it does claim are real.

5. Comfort, accessibility and inclusion

- The site is level, so the whole park is step-free. The only change of level in the scheme is the Oasis Basin, which is ramped.
- Every room is reached from the shaded primary route by a single alley — no room requires crossing another to arrive.
- Shaded rest (the majlis pods) is never far from the walk, so a user who needs to stop can always reach shade.
- 6 drinking fountains and 2 universally accessible restroom blocks are distributed against actual dwell points, not spaced evenly for the drawing.
- Play is divided by age group, with family seating in shade and clear sightlines to both zones.

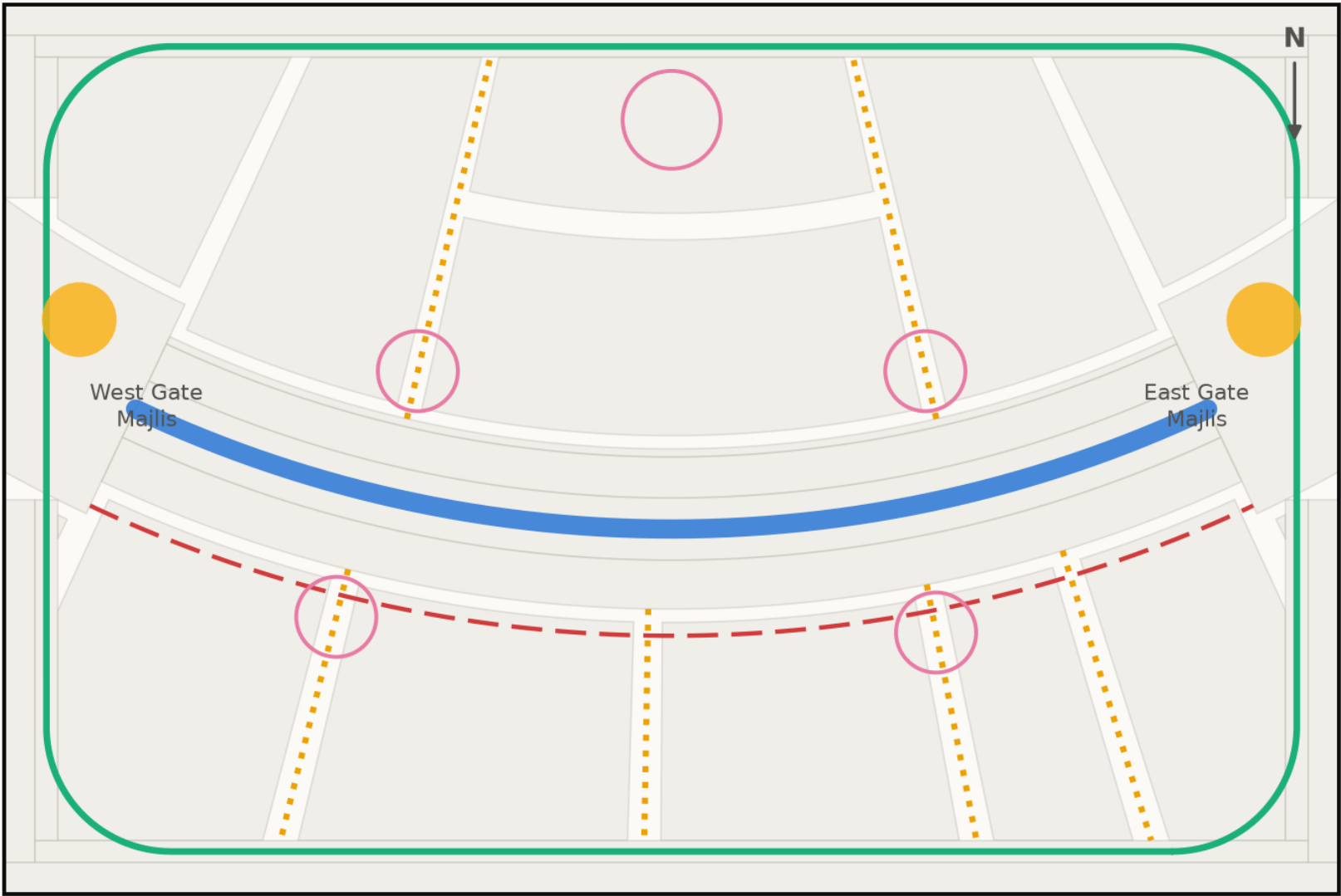
6. Day and night

The brief asks for a park that works across the whole day. The convex face carries the uses that run after dark — the plaza, the event lawn and the souk — because its exposure stops being a liability once the sun is down. The crescent is lit from within so the arc reads at night as it does by day, and Al Madar is lit at ankle height for early and late running.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



- Al Mamsha — primary route, shaded, step-free
- Al Sikkak — shaded alleys to the rooms
- Al Madar — 438 m running loop
- Service / emergency vehicle access
- Majlis — shaded rest, never more than 33 m from the walk

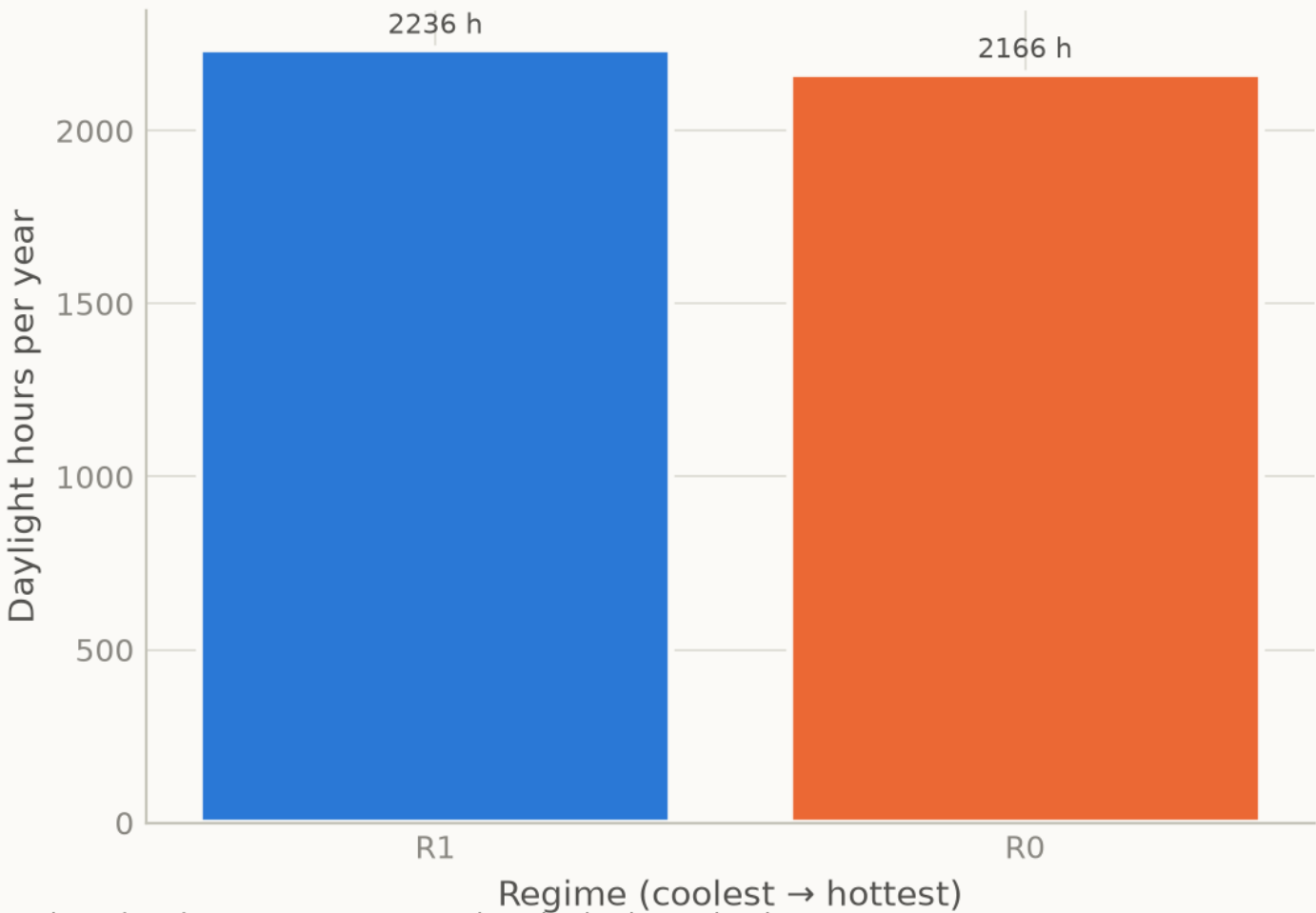
Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: src/plan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig08 microclimate regimes

Analysis output — computed from project data.

The park's 2 operating regimes, found by clustering

k selected by silhouette score, not chosen — daylight hours only



Each regime becomes a programming slot in the activation strategy.

Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

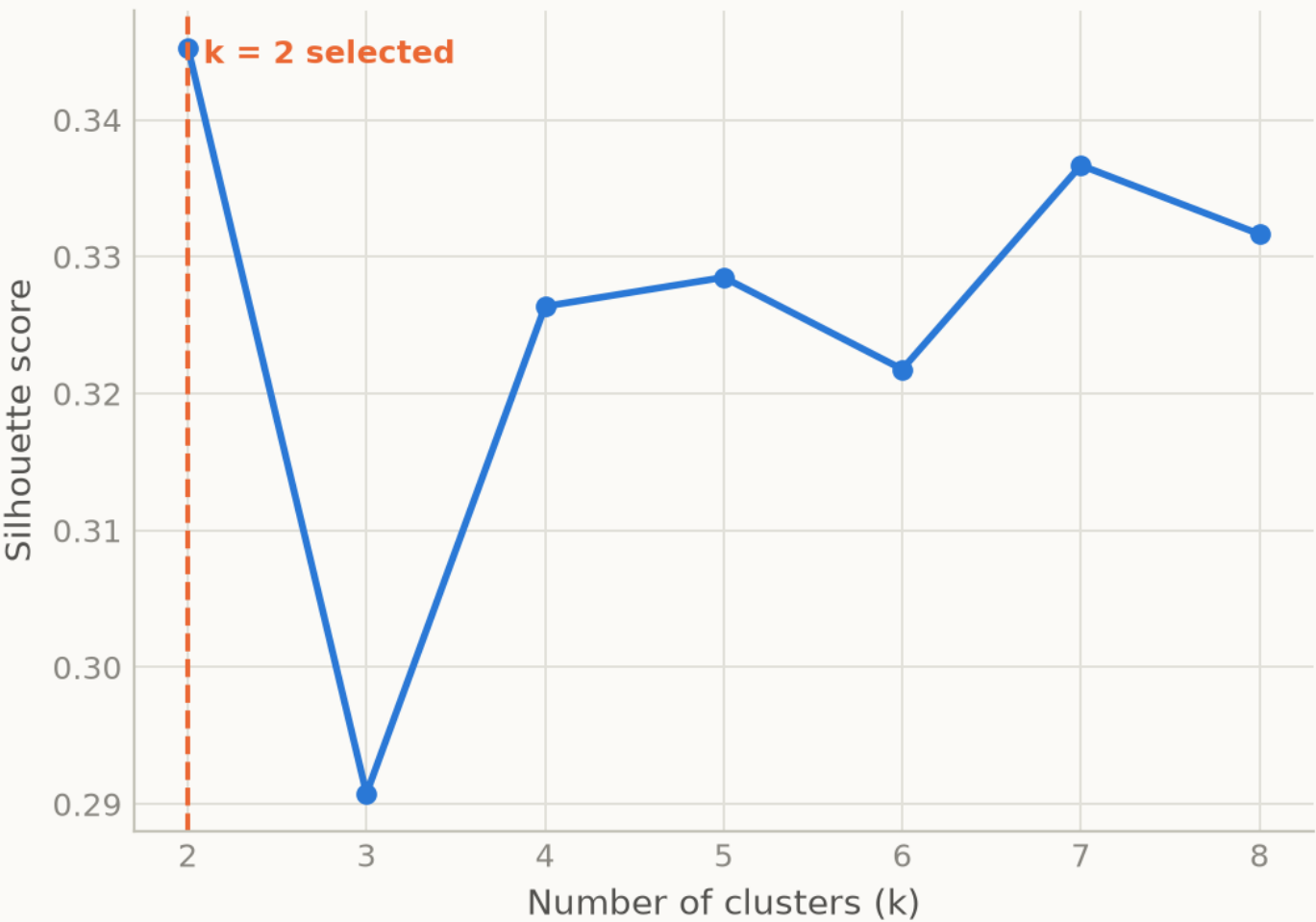
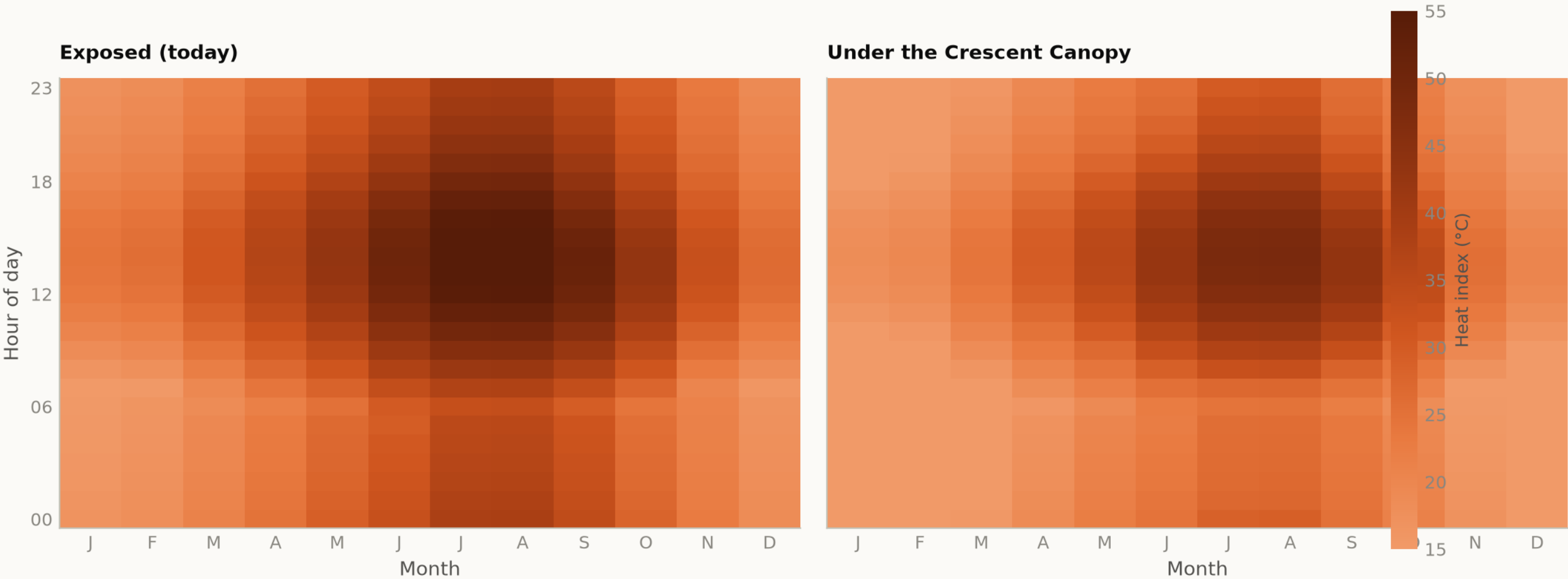


Fig09 diurnal comfort

Analysis output — computed from project data.

When is the park comfortable?

Mean heat index by hour and month — exposed today, shaded as designed



The design opens up the late afternoon in spring and autumn, which is when the catchment most wants to use the park.
Source: NCM Dubai climate normals 1977-2015 (39-year record) — hourly series MODELLED, see src/climate.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The complete project, and where to check it

Every link below is live. The repository holds the data, the code, the trained models and the tests, and runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/

START HERE

EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language
PROJECT_PLAN.md	Requirements, phases, status, what is left
README.md	The design argument, written for a juror
notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded

THE GEOMETRY AND THE MODELS

src/plan.py	Single source of the crescent geometry
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/solar.py	Sun position and shadow ray-tracing
src/dataset.py	Assembles the ML training tables
src/models.py	The four models
src/costing.py	The cost model against the AED 35 M ceiling

THE DATA, AS ISSUED AND AS PROCESSED

data/raw/sources.json	Every source dataset, with its period
DATA_SOURCES.md	Sources and their stated limitations
data/raw/site_zoning_schedule.csv	The measured room schedule
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/cost_plan.csv	The capital cost plan, line by line

THE RESULTS, AND THE CHECKS ON THEM

models/model_metrics.json	Trained-model metrics
models/headline_metrics.json	The headline numbers
tests/test_pipeline.py	38 correctness checks
archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why

THE DRAWINGS

figures/fig10_masterplan.png	Masterplan and room schedule
design/visuals/section_crescent.png	Section A–A at midspan
design/visuals/circulation_crescent.png	Circulation and accessibility
design/visuals/facilities_crescent.png	Commercial & Service Facilities Map
design/visuals/planting_crescent.png	Planting plan, 131 trees

Links resolve once the repository is published. GitHub renders PDFs and notebooks in the browser — nothing needs to be downloaded or cloned.